3) $F_c = \frac{mv^2}{R}$, $P = mg$

$mv^2 \geq mg$ $\quad v \geq \sqrt{gR}$

$E_p = E_c$

$E_p = mgx$, $E_c = \frac{mv^2}{2}$

$mgx = \frac{mv^2}{2}$

$x = \frac{v^2}{2g}$

$x = \frac{(\sqrt{gR})^2}{2g} = \frac{gR}{2g}$

$x = \frac{R}{2}$ $\quad h = x + 2R$

$h = \frac{R}{2} + 2R = \frac{R + 4R}{2}$

$h = \frac{5R}{2} = \boxed{2{,}5R}$

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